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Lecture 5: Design documents and pre-analysis plans

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Pre-registration repositories

- Open Science (OSF) https://osf.io/registries also pre-prints
- (absorbed Evidence in Governance and Politics (EGAP) https://osf.io/registries/egap/discover)
- AsPredicted https://aspredicted.org/
- American Economic Association https://www.socialscienceregistry.org/

Making research transparent

- Replication repository
  - Harvard dataverse https://dataverse.harvard.edu/
  - GitHub
- Pre-print servers
  - Also on OSF https://osf.io/preprints/
  - https://arxiv.org/
  - https://www.ssrn.com/

Overview

- Why pre-register?
- The role of discovery in pre-registered trials
  - Confirmatory vs. exploratory research.

(Potential) research timeline

- Project planning (/ funding?)
- IRB protocol approval
- pre-registration/design document?
- Study implementation/ data collection
- Updates to pre-registration?
- Analysis
- Registered report write-up?
- Paper write-up
- Pre-print
- Journal revisions and publication
- Finalize and post replication files

Example on OSF

- Appeals to Candidacy: Voice and Pragmatism: https://osf.io/gj9vh/

What to include in your pre-registrations

- Research question and hypothesis
  - How does this project contribute to greater scientific knowledge?
  - What are you expecting to find?
- (light) Literature review
- Study population / procedure for obtaining sample frame. Inclusion criteria?
  - Further information about context?
- Is informed consent obtained? If not, why not?
- What is being manipulated? Be specific.
- Randomization procedures/unit of randomization.
- Data collection: how/when is data collected? What is the unit of measurement? (individual? village? school? multiple of these?)
- What outcomes and covariates are measured? How are they measured: surveys, behavioral games, behavioral measures, administrative data? Any justifications for measurement choices.
- Analyses/statistical tests? (Just a general idea, but describe e.g., any subgroup analyses or tests for heterogeneous treatment effects. Which groups are you interested in, and what do you expect to find?)
- Ethical considerations

Other considerations

- IRB protocol vs. pre-registration
- Deviating from pre-registration
- Pre-registration for observational studies

Ethical considerations

- When is it ethical to conduct research on subjects who are not able to give informed consent?
- Under what conditions is risk acceptable?
- Under what conditions should a research project be aborted?
- When is deception justifiable?

Additional reading

- "More and more scientists are preregistering their studies. Should you?" Science. 21 Sep 2018. https://www.science.org/content/article/more-and-more-scientists-are-preregistering-their-studies-should-you
- "The preregistration revolution." PNAS. 12 Mar 2018. https://doi.org/10.1073/pnas.1708274114
- OSF templates https://osf.io/zab38/wiki/home/